

LALITHA PRASAD PALNATI

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PROFESSIONAL SUMMARY

Motivated fresher DevOps & Cloud Engineer with strong hands-on project experience in end-to-end CI/CD pipeline design, cloud infrastructure automation, container orchestration, and DevSecOps practices. Skilled in Kubernetes, Docker, Terraform, Azure, AWS, and observability tools (Prometheus & Grafana). Demonstrated ability to build, secure, and monitor production-grade infrastructure. B.Tech (ECE) graduate with a practical mindset, eager to contribute to dynamic engineering teams across cloud, DevOps, or SRE roles.

TECHNICAL SKILLS

- **Cloud Platforms:** Azure (VM, VNet, ACR, AKS, VMSS, Key Vault, Bastion, Monitor, Storage), AWS (EC2, VPC, Security Groups, S3)
- **CI/CD & Automation:** Azure DevOps (YAML Pipelines), Jenkins, Git, GitHub — multi-stage automated build, test, security, and deploy pipelines
- **Infrastructure as Code:** Terraform — infrastructure provisioning and state management on Azure & AWS; modular, reusable IaC configurations
- **Container Orchestration:** Kubernetes (AKS) — deployment manifests, rolling updates, liveness/readiness probes, scaling, namespace management
- **Containerization:** Docker — image building, multi-stage Dockerfiles, container registry (ACR) push/pull, vulnerability scanning with Trivy
- **Monitoring & Observability:** Prometheus, Grafana — metrics collection, alerting rules, SLI/SLO dashboards; Azure Monitor for cloud-level observability
- **Security & DevSecOps:** SonarQube (static code analysis), Trivy (container security scanning), Azure RBAC, Managed Identities, NSG policies
- **Operating Systems & Scripting:** Linux (Ubuntu) — system administration, Bash scripting, process/service management, SSH, networking
- **Development & Build Tools:** Python (Flask), Java (Maven), MySQL — used in project development and deployment pipelines
- **Communication Protocols:** TCP/IP, HTTP/HTTPS, SSH, DNS — applied across cloud deployments, microservices networking, and API communications

PROJECTS

Enterprise-Grade CloudShield DevSecOps Pipeline on Azure

Tools: Azure (VM · VNet · ACR · AKS · Storage) · Terraform · Azure DevOps YAML Pipelines · Git · Docker · Kubernetes · Java/Maven · SonarQube · Trivy · Prometheus · Grafana · Linux

- Designed and provisioned complete Azure cloud infrastructure from scratch using Terraform IaC — including VNet, Subnets, VM, Storage Account, ACR, and AKS — following infrastructure-as-code best practices.
- Built a multi-stage CI/CD pipeline (YAML) on Azure DevOps automating the full lifecycle: code checkout → Maven build → SonarQube code quality gate → Trivy container scan → Docker image build → ACR push → AKS deployment.
- Configured a self-hosted Azure DevOps agent on an Ubuntu Linux VM; managed SSH, system services, and environment variables for reliable pipeline execution.
- Implemented Kubernetes rolling deployments with liveness and readiness health checks on AKS, ensuring zero-downtime releases and automated recovery from pod failures.
- Enforced DevSecOps practices by integrating SonarQube (static code analysis) and Trivy (CVE scanning) as pipeline gates — blocking deployments with critical vulnerabilities.
- Set up Prometheus metric scraping and Grafana dashboards for real-time observability of application health, infrastructure performance, and alerting on threshold breaches.
- Outcome: End-to-end automated, secure, and observable deployment pipeline — awarded 3rd Prize at CloudShield DevSecOps Hackathon 2026, V Cube Software Solutions.

Deployment and Management of Full-Stack Flask Application with MySQL

Tools: AWS (EC2 · VPC · Security Groups · Network ACLs) · Python/Flask · MySQL · Linux (Ubuntu) · pip · Virtual Environments

- Architected and deployed a production-style 3-tier application on AWS with separate EC2 instances for Presentation (Web), Application (Flask), and Data (MySQL) layers, demonstrating tier isolation and scalability.
- Configured AWS VPC networking, Security Groups, and Network ACLs to enforce least-privilege communication between tiers, simulating enterprise-grade network security.
- Administered Linux servers across all three EC2 instances — installed Python dependencies using pip and virtual environments, configured systemd services, and managed application startup/restart automation.
- Managed MySQL database operations including schema creation, user privilege management, and secure backend-to-database connectivity over private subnets.
- Gained hands-on experience in AWS infrastructure operations, multi-tier application reliability, Linux server administration, and secure cloud networking.

EDUCATION

B.Tech – Electronics & Communication Engineering | Eswar College of Engineering (JNTU Kakinada) | CGPA: 8.06 2025

Intermediate (MPC) | AP Model Junior College | 85.3% 2021

SSC (10th) | Sri Viswasanthi High School | GPA: 9.7 2019

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Azure Certification Training – V Cube Software Solutions, April 2025.
- 3rd Prize – CloudShield DevSecOps Hackathon 2026, V Cube Software Solutions (competitive national-level hackathon).
- 2nd Prize – College Engineering Day, Technical Quiz & Innovation category.